Network Intrusion Detection System

Channel Islands
CALIFORNIA STATE UNIVERSITY

Luis Vazquez • Dr. Vida Vakilian • COMP 499

Background

NETIDS is a Python-based Network Intrusion Detection System (NIDS) developed as a senior capstone project. Traditional enterprise IDS tools are expensive. NETIDS provides a fully open-source, transparent, and extensible alternative for learning and small-scale network security monitoring.

Motivated by growing interest in cybersecurity and hands on threat detection, NETIDS was designed to run on commodity hardware and virtual environment, making it accessible to students and home-lab enthusiasts alike. The project integrates real-time packet analysis with AI-assisted SOC reporting via a locally-hosted LLM

Overview

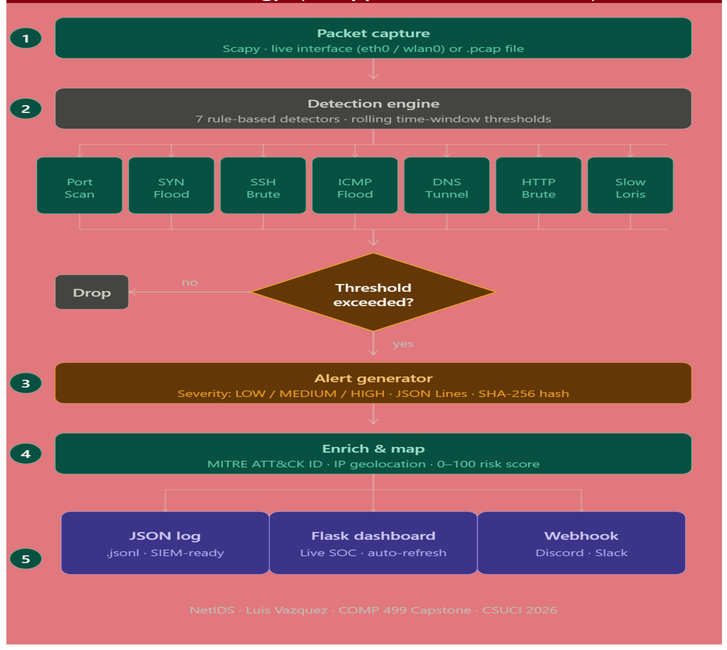
Problem Statement :

Most network threats go undetected on small networks due to lack of affordable monitoring tools. NETIDS fills this gap with a lightweight, AI-augmented detection Engine

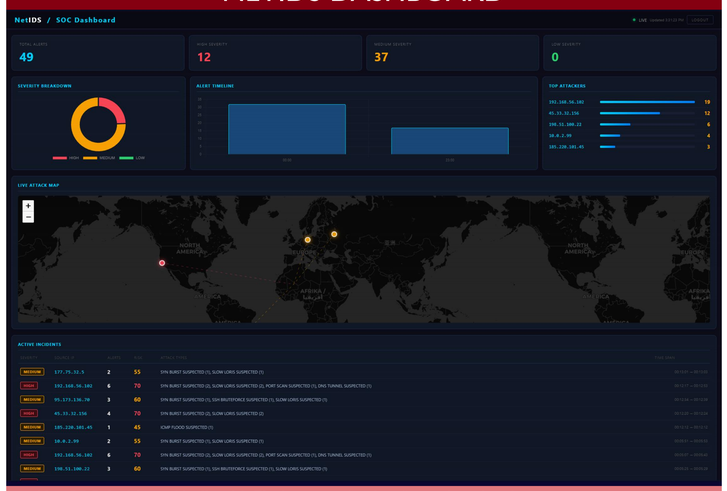
Key Features

- Real-time packet capture (Scapy) and offline PCAP file analysis
- 7 detectors: port scan, SYN flood, SSH brute force, ICMP flood, DNS tunneling, HTTP brute force, Slow Loris
- Three-tier severity per detector (LOW / MEDIUM / HIGH) with configurable thresholds
- MITRE ATT&CK technique mapping on every alert
- IP geolocation and ASN enrichment via ip-api.com
- Incident correlation with 0–100 risk scoring and attack chain detection
- Colored terminal output + JSON Lines alert log (SIEM-ready)
- Flask web dashboard with live incident view and timeline
- Webhook notifications (Discord / Slack) and optional iptables auto-blocking
- Runs on Ubuntu VM / bare-metal (root required for live capture)

Methodology (or type a new title here)

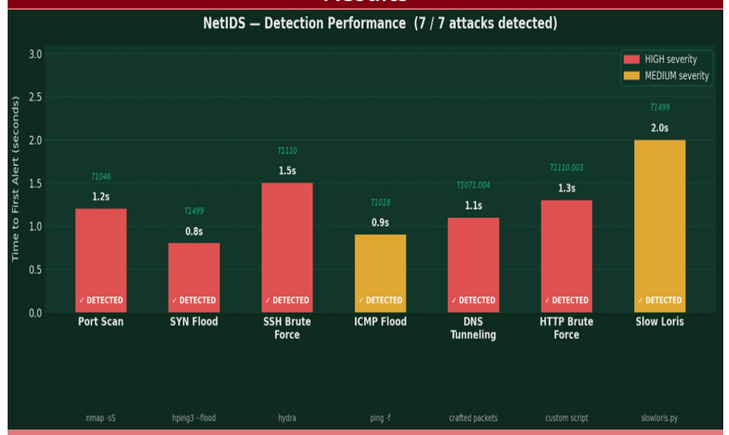


NETIDS DASHBOARD



- Live SOC dashboard built with Flask, auto-refreshes from alerts.jsonl
- Summary counters: total alerts, HIGH / MEDIUM / LOW severity counts at a glance
- Severity breakdown donut chart and hourly alert timeline (last 12 hours)
- World map plots geo-located attack origins, color-coded by severity
- Top 5 attacker IPs ranked by alert volume
- Active incidents table source IP, risk score (0–100), alert types, attack chain, first/last seen timestamps
- Password-protected login with rate limiting (max 5 attempts per 60s)
- REST API endpoints (/api/alerts, /api/stats, /api/geo, /api/incidents) support filtering by severity, type, and source IP

Results



- All 7 attack types successfully detected in live capture mode
- Average time to first alert: 1.3 seconds from first malicious packet
- 5 of 7 attacks classified as HIGH severity, 2 as MEDIUM
- Zero false positives observed during normal browsing baseline test
- Detection triggered within a single rolling time window (8–30 seconds depending on attack type)

Conclusion

Conclusion:

- NetIDS successfully detects 7 attack types: port scan, SYN flood, SSH brute force, ICMP flood, DNS tunneling, HTTP brute force, and Slow Loris
- Runs entirely on commodity hardware with no cloud dependency or licensing cost
- Real-time alerting with average sub-2 second latency from first malicious packet
- Live Flask SOC dashboard provides instant visibility into active incidents, risk scores, and attack origins

Future Work:

- Add ML-based anomaly detection using the isolation forest model already in the repo
- Integrate TLS fingerprinting (JA3/JA4) to detect threats in encrypted traffic
- Package as Docker container for easy deployment
- Extend Windows support via Npcap

References

- Scapy Documentation. "Scapy: Packet Manipulation Tool." https://scapy.net/
- MITRE ATT&CK. "ATT&CK Framework." https://attack.mitre.org/
- Snort IDS. "Snort User Manual." https://www.snort.org/
- Suricata. "Suricata Documentation." https://suricata.io/
- Wireshark Foundation. "Wireshark User Guide." https://www.wireshark.org/docs/

Technologies Used

Python logo — © Python Software Foundation
GitHub logo — © GitHub, Inc.
Visual Studio Code logo — © Microsoft
Ubuntu logo — © Canonical Ltd.

Special thanks

Special thanks to my Capstone Advisor for the guidance in this project